

1. `@SpringBootApplication` : is added on the main class , in spring boot application , to run it as a standalone JAR / WAR

`@SpringBootApplication` =

1.1 `@Configuration` – Designates this main class as a configuration class for Java configuration. In addition to beans configured via component scanning, an application may want to configure some additional beans via the `@Bean` annotation as demonstrated here. Thus, the return value of methods having the `@Bean` annotation in this class are registered as beans.

1.2 `@EnableAutoConfiguration` – This enables Spring Boot's autoconfiguration mechanism. Auto-configuration refers to creating beans automatically by scanning the classpath.

1.3 `@ComponentScan` – Typically, in a Spring application, annotations like `@Component`, `@Configuration`, `@Service`, `@Repository` are specified on classes to mark them as Spring beans. The `@ComponentScan` annotation basically tells Spring Boot to scan the current package and its sub-packages in order to identify annotated classes and configure them as Spring beans. Thus, it treats the current package as the root package for component scanning.

Annotation added automatically :

`@EnableWebMvc`: Marks the application as a web application and activates setting up a `DispatcherServlet` , `HandlerMapping` , `ViewResolvers` . Spring Boot adds it automatically when it sees `spring-webmvc` on the classpath.

2. How to use the `@SpringBootApplication` annotation

In order to run a Spring Boot application, it needs to have a class with the `@SpringBootApplication` annotation.

eg :

```
package com.app;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication
public class Main {
    public static void main(String[] args) {
        SpringApplication.run(Main.class, args);
    }
}
```

The Main class has the `@SpringBootApplication` annotation

It simply invokes the `SpringApplication.run` method.

This starts the Spring application as a standalone application, runs the embedded servers and loads the beans.

Normally, such a main class is placed in a root package above other packages. This enables component scanning to scan all the sub-packages for beans.